

ML Specification Report — afsis-soil-properties

Africa Soil Property Prediction Challenge · our from-scratch agent (kernel-methods spectral pool + tree-search v3)

Figures grounded in the competition's `config.yaml`, `dossier.json`, `STATUS.md`, `experiments.json`, `experiments_tree_v3.json`, `experiments_tree_v3_run1_diluted_blend.json`, and `scripts/state/final.json`, with leaderboard and benchmark figures from the study's frozen three-way score table.

Overview

The task is to predict five physical and chemical soil properties — Ca, P, pH, SOC, Sand — from mid-infrared spectral measurements of African soil samples (metric: MCRMSE, the mean columnwise RMSE over the five targets, minimize). We solve it with a from-scratch pipeline built for the $p \gg n$ regime: a pool of kernel ridge / SVR models over five spectral preprocessing variants (plus PLS and LightGBM-on-PCA for diversity), validated by a nested GroupKFold over hidden site groups and refined by an 80-node tree search. Our champion, tree-search node #73 — a per-target greedy (Caruana forward-selection-with-replacement) blend of all 64 solo nodes — scores **honest CV MCRMSE 0.444076** (0.428772 when the blend weights are fit in-sample) and, on the real leaderboard, **Public 0.45207 / Private 0.49517**, ranking **659/1233 (percentile 46.6)**. Among the three benchmark agents ours is the **leaderboard winner**: AIDE posts the best local CV (0.40664) but finishes behind us on both boards (Private 0.49861, percentile 45.6), and NVIDIA's kernel reproduction trails far behind (Private 0.69855, percentile 18.8).

Why it matters. Sub-Saharan African soils are chronically under-surveyed because wet-chemistry assays are slow and expensive; predicting nutrient status from cheap infrared spectroscopy enables soil mapping at continental scale, informing fertilizer recommendations, land management, and food-security policy.

Data

Purpose of Data. Predict five continuous soil-property targets per sample — a spectral multi-target tabular regression in a $p \gg n$ regime (the dossier's task family). **Data Format** is **tabular CSV** dominated by an ordered signal: **3,578 spectral columns** (mid-infrared absorbance, wavenumbers `m599.76` to `m7497.96`), **15 spatial covariates** (BSAN...TMFI — anonymized satellite/terrain rasters), and **1 categorical** (`Depth`: Topsoil/Subsoil), 3,594 features in all. **Data Volume** is **1,157 training rows / 727 test rows** — tiny n against huge p , which drives every modelling choice below.

Data Quality is high on the surface — **zero nulls**, all features pre-standardized — but the structure hides two traps. The targets are pre-standardized yet heavily right-skewed (skew P 7.45, Ca 4.71, SOC 2.45, pH 0.87, Sand 0.39), so squared error is driven by a handful of extreme rows. And the rows are not i.i.d.: the strongest EDA signals were

- **580 distinct spatial signatures across 1,157 train rows** — 565 sites each contribute a Topsoil/Subsoil pair, so a plain shuffled KFold puts two near-duplicate samples of the same site in different folds and leaks location.
- **Zero train ↔ test site overlap**, and test locations sit systematically far from train in the 15-d covariate space (train → train nearest-neighbour distance median 0.306 vs test → train median 0.956; 46.7% of test locations exceed the train 95th percentile) — test is drawn from held-out landscapes, not the same clusters.
- **Kernelization is worth ~0.032 MCRMSE** over linear ridge (0.475432 vs 0.507098) — the spectra → property map is nonlinear.
- The spatial covariates alone are weak (KRR-RBF on spatial only: 0.726751) — the spectra carry the signal.
- **P is by far the hardest target**: honest per-target RMSE 0.917442, versus 0.321491 (Ca), 0.338487 (pH), 0.321772 (SOC), 0.321187 (Sand) — one column dominates the achievable headroom, motivating per-target models and per-target blend weights.

Annotation Guidelines. The labels are five continuous, pre-standardized laboratory soil measurements (no de-standardization key is provided); submissions are **real-valued predictions for all five columns**, scored by MCRMSE — an L2-family metric that wants conditional means, not classes.

Feature Set. The saturated **CO2 band (2352–2380 cm⁻¹, 15 columns) is dropped**; the remaining spectra are served in **five preprocessing variants** — raw, SNV, SG-d1+SNV, SG-d2+SNV, SG-d1(w11)+SNV (Savitzky–Golay derivatives + standard-normal-variate) — each solo model picking one variant, plus z-scored spatial covariates (with a tunable `spatial_weight` knob) and binary `Depth`.

Splitting strategy. The load-bearing decision. Validation is a **nested GroupKFold(5)** (seed = 42) with the group key = the 580 spatial-signature site groups recovered from the 15-covariate tuples, so a site's Topsoil/Subsoil pair can never straddle folds — mirroring the held-out-landscape train → test relation. Each target's ridge alpha is selected by an **inner GroupKFold(4) that only ever sees outer-train rows**, making every solo score honest by construction. Blend scores are honest only after the weights are **re-fit leave-one-outer-fold-out**: the champion reads 0.428772 with in-sample weights but **0.444076** honestly — a measured weight-fitting optimism of **+0.015304**, squarely inside the +0.0146...+0.0186 band the experience library predicted.

Models & Architecture

Purpose of Architecture. A five-target soil-property regressor that minimizes MCRMSE under a held-out-landscape split. **Architecture Type** is a **regularized kernel-methods ensemble** — kernel ridge regression (RBF, polynomial-2, Laplacian kernels) and SVR-RBF as the core learners, chosen over GBDT for the $p \gg n$ regime, with PLS regression and LightGBM-on-PCA as structurally different decorrelating members. The champion is a **per-target greedy blend of 64 solo nodes** discovered at tree-search node #73.

Each member takes the same **Input Format** — one spectral preprocessing variant concatenated with the weighted spatial covariates and Depth — a 1-D row vector per sample (**Input Dimension** $\approx$ 3,563 spectral columns after the CO2-band drop, + 15 spatial + Depth; the spectra are treated as an ordered signal at preprocessing time, then flattened for the kernel).

Architecture Description. Each kernel member fits five independent per-target regressions (MCRMSE weights columns equally and difficulty varies wildly, so no multi-output head). **Model Complexity** is governed by the kernel matrix, $n \times n = 1,157^2$, not a parameter count — regularization (per-target alpha, kernel width, `spatial_weight`) does the work. The best solo is **node #42, KRR-RBF on SNV spectra with `spatial_weight` 0.25, honest 0.469086**; the blend layer on top assigns per-target weights by Caruana greedy forward selection (60 rounds) over out-of-fold predictions. Per-target weights matter a lot: 0.433071 with per-target weights vs 0.452840 with one shared weight vector over the same 16 members.

Training procedures

Training proceeds as the standard ladder — naive floor, linear protocol, then tree search — every score on the same nested-GroupKFold scale:

Stage	Configuration	MCRMSE
naive floor	per-fold target mean	1.024945
linear baseline	Ridge (linear kernel) on SG-d1 spectra + spatial	0.507098
solo pool (16 configs)	best = SVR-RBF on SG-d1 + spatial (tree root)	0.471939
linear-stage blend	15-member per-target Dirichlet + coordinate ascent (weights in-sample)	0.433219
tree search v3 (node #73)	64-member per-target greedy kitchen-sink blend (weights in-sample)	0.428772
champion, honest	same blend, weights re-fit leave-one-outer-fold-out	0.444076

The tree search (`harness_v3`) evaluated **80 nodes (67 solo / 13 blend) in 74 s**, first beating the linear-stage best at **evaluation 17**. The budget was raised from the default 60 to 80 because 15 of the first 16 nodes are zero-compute cached-OOB re-imports of the linear pool; the search stopped on the **hard budget cap (80/80)** with the mandatory explore burst (injected at `n_eval = 70`) still improving — the burst’s kitchen-sink blend is the champion. The sanity gate fired once (burst seed #71, linear kernel on raw spectra, 0.686914 against a bound of 0.549675 — lineage plateaued immediately); the cost guard never fired; 19 lineages plateaued and 12 dedup rejections were logged.

The one real finding is a negative result. The first tree-search run re-seeded only 4 of the 16 linear-pool solos, and its kitchen-sink blend scored **0.449717 — worse than the linear blend it was meant to beat**. Re-seeding the entire linear pool as first-generation lineages, with no other change, moved the champion to 0.428772. The weight-search method is second-order: at 64 members greedy 0.428772, Dirichlet 0.428899, bagged greedy 0.430371 — **pool diversity, not weight search, explains essentially the whole gap** (run-1 tree archived as `experiments_tree_v3_run1_diluted_blend.json`, experiment #6).

The **Loss Function** family is L2 throughout, per the dossier’s metric trap (“RMSE family, not MAE”) — kernel ridge minimizes squared error in closed form; the SVR members’ ϵ -insensitive loss is admitted for diversity, with model selection always on MCRMSE. The **Optimization Algorithm** is closed-form kernel solves / quadratic programming — **not** SGD/ADAM — with blend weights by greedy forward selection. There is **no Learning Rate** and **no Scheduler** (*N/A — closed-form and QP fits, no gradient-descent schedule*; the lone LightGBM-on-PCA member uses standard boosting shrinkage) and **no Batch Size** (*N/A — full-matrix kernel fits, not mini-batched*).

Training Duration for the tree-search stage was **74 s of wall-clock over 80 evaluated nodes** (kernel fits at $n = 1,157$ are cheap; no blend approached the 45 s cost-guard threshold); linear-stage and total pipeline wall-clock were not recorded. **Training Memory** was not explicitly capped — the dominant object is the $1,157^2$ kernel matrix, trivial on the arm64 CPU machine, peak not recorded. **Transfer Learning** is *N/A (no pretrained weights)*; its analogue is **cross-competition experience injection** — the TASK-SPECTRAL and TASK-BLEND-DECOR priors and the $+0.0146\dots+0.0186$ weight-optimism band carried in from the experience library and verified here. **Data Augmentation** is *N/A (tabular/spectral)*; the analogue is the five spectral preprocessing variants, which manufacture the view diversity the blend feeds on.

Reproducibility Standards: fixed seed 42 for the fold construction, per-target alphas selected deterministically inside the nested inner loop, and the tree root re-imports the linear-stage cached OOF **digit-verified** against the original run; the full backtrack log, dedup log, and the negative-result run-1 tree are all preserved in the workspace JSONs.

Inference procedures

Decision Threshold is *N/A* — MCRMSE is a regression metric, so we submit raw real-valued predictions for all five targets and never threshold. No post-processing is applied. The submission (`submission.csv` , 727 rows) was verified column-for-column and PIDN-order-identical to `sample_submission.csv` . **Inference Duration** was not separately profiled — kernel scoring of 727 test rows against 1,157 support rows is sub-second per member — and **Inference Memory** is negligible (not capped).

Evaluation & Benchmarking

Task for Performance Evaluation. Predict Ca, P, pH, SOC, and Sand for the 727 held-out-landscape test samples. **Performance Metrics.** MCRMSE (minimize) is the sole competition metric; our champion's honest local number is **CV MCRMSE 0.444076**, improving $1.024945 \rightarrow 0.507098 \rightarrow 0.471939 \rightarrow 0.433219$ (in-sample) $\rightarrow 0.428772$ (in-sample) down the ladder, with the honest leave-fold-out re-fit as the headline. On the real leaderboard the same submission scores **Public 0.45207 / Private 0.49517**, rank **659/1233 (percentile 46.6)** — and the honest grouped-CV number (0.444076) tracks the public score (0.45207) closely, vindicating the nested-GroupKFold design. The champion's 0.000127 margin over the runner-up blend is noise; the gaps to the 16-member blends (~ 0.0043) and to the best solo (~ 0.040) are not.

Performance Benchmarking. Per the benchmark set's standing convention, **between-agent claims use the leaderboard numbers**; each agent's local CV is its own internal view under its own scheme (grouped and nested for us, unknown and not comparable for the others) and is **never used for cross-agent ranking**. The frozen three-way table:

Agent	Approach	Local CV	Public LB	Private LB	Percentile
Our agent	from-scratch kernel pool + tree-search v3	0.444076	0.45207	0.49517	46.6
AIDE	open-source AIDE agent	0.40664	0.45636	0.49861	45.6
NVIDIA	reproduces a public kernel	0.48493	0.61276	0.69855	18.8

Our agent wins on the leaderboard (`lb_winner` = mine): Private 0.49517 vs AIDE’s 0.49861 and NVIDIA’s 0.69855, and the same ordering holds on the public board. AIDE is the `local_winner` (0.40664) — but its local edge does not survive contact with the private test, while our honest grouped CV sits much nearer its own leaderboard outcome; the site-group split policy that made our local number *look* worse is exactly what made it *transfer*. NVIDIA’s copied kernel collapses on this comp (percentile 18.8), the recurring failure mode of vote-based kernel selection on older research competitions.

Fields marked “not recorded”: linear-stage and total pipeline wall-clock; training peak memory; inference duration and memory; target physical units and standardization constants (not provided in the data). STATUS.md was written before benchmark scoring and marks the run OOF-only; the leaderboard figures above come from the study’s frozen three-way score table.